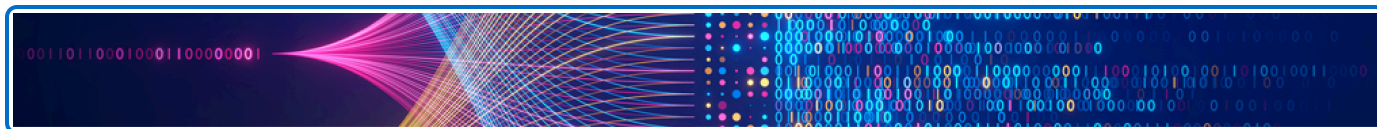


Week 1 Overview



Week 1 Overview

Welcome to our first lesson on neural networks, where we explore the foundational concepts of deep learning. This week, we'll explore how networks are built from logistic regression neurons using activation functions like sigmoid, tanh, or ReLU. Neural networks consist of layers of neurons to transform an input vector into an output vector in stages. The design of the output layer determines whether a network solves a regression or a binary classification task.

Week 1 Introduction

(00:55)



Learning Objectives

At the end of this week, you should be able to:

- Explain how neurons implement logistic regression using activation functions (e.g., sigmoid, tanh, or ReLU).
- Describe the architecture of neural networks by detailing the roles of input, hidden, and output layers.
- Illustrate the representation of a neural network layer using linear algebra notation.
- Analyze how the output layer's configuration dictates whether the network performs regression or binary classification.
- Set up and implement basic feed-forward networks for regression and binary classification in Keras.

This Week's Activities

Assignment	Title	Type	Graded / Not Graded	Due
Knowledge Check	Week 1 Knowledge Check	Multiple choice	Ungraded	Sunday
Assignment	Coding Homework	Jupyter Notebook	Graded	Sunday

Weekly Cadence

- Faculty Live Session: Tuesday 7-8 pm ET (day change due to U.S. holiday on Monday)
- Weekly deadline: Sunday, 11:59pm ET